

CAPSTONE PROJECT

Railway Management System

Real-Life Example: Indian Railways Passenger Reservation

1. Introduction

Railway transportation handles a large amount of information related to trains, passengers, stations, reservations, seats and payments. Managing this information manually can lead to duplication, errors and difficulty in retrieving records.

A Railway Management System is a database-based system that manages passenger reservations and railway-related information efficiently.

For this capstone, we consider a real-life scenario similar to an online Indian railway passenger reservation system. A passenger searches for a train, selects a journey, reserves a seat and makes a payment. The system stores all these details in related database tables.

2. Real-Life Scenario

Consider a passenger named Rahul Sharma who wants to travel from Delhi to Mumbai.

- Passenger → Searches for Train → Selects Train → Makes Reservation → Seat is Assigned → Makes Payment → Reservation Confirmed

3. Problem Statement

The system should maintain information about passengers, trains, stations, reservations and payments.

- Store passenger information.
- Store train information.
- Store station information.
- Register passenger reservations.
- Assign seats to passengers.
- Record payments.
- Track reservation status.
- Retrieve passenger booking details.
- Retrieve train and station information.
- Maintain relationships between different entities.
- Prevent invalid or inconsistent data.

4. Objectives

- Maintain passenger records efficiently.
- Maintain train and station information.

- Manage railway reservations.
- Store payment information.
- Establish relationships between related entities.
- Retrieve information using SQL queries.
- Use joins for combining information from different tables.
- Use nested and complex queries for advanced data retrieval.
- Use views for simplified access to booking information.
- Use triggers for automatic database operations.
- Maintain data integrity using constraints.

5. Scope

Passenger Management

- Passenger ID
- Name
- Age
- Gender
- Phone number

Train Management

- Train ID
- Train name
- Source
- Destination
- Departure time
- Arrival time

Station Management

- Station ID
- Station name
- Location

Reservation Management

- Reservation ID
- Passenger ID
- Train ID
- Payment ID
- Seat number
- Journey date
- Reservation status

Payment Management

- Payment ID
- Amount
- Payment mode
- Payment status

6. Database Design

The Railway Management System contains the following main tables and relationships:

Table	Primary Key	Important Foreign Keys
Passenger	Passenger_ID	—
Train	Train_ID	—
Station	Station_ID	—
Payment	Payment_ID	—
Reservation	Reservation_ID	Passenger_ID, Train_ID, Payment_ID
Train_Station	Train_ID + Station_ID	Train_ID, Station_ID

Relationship structure: Passenger → Reservation → Train; Reservation → Payment; Train → Train_Station → Station.

7. Real-Life Example in Database

Suppose Rahul Sharma books Rajdhani Express.

Entity	Example Data
Passenger	Passenger_ID = 101; Name = Rahul Sharma; Age = 21; Gender = Male
Train	Train_ID = 1001; Train_Name = Rajdhani Express; Source = Delhi; Destination = Mumbai
Reservation	Reservation_ID = 401; Passenger_ID = 101; Train_ID = 1001; Seat_Number = A1-10; Journey_Date = 15-09-2026; Status = Confirmed
Payment	Payment_ID = 301; Amount = ₹1500; Payment_Mode = UPI; Payment_Status = Paid

8. SQL Operations Implemented

DDL and DML: Used to create and populate Passenger, Train, Station, Reservation, Payment and Train_Station tables.

Integrity Constraints: PRIMARY KEY, FOREIGN KEY, NOT NULL, UNIQUE, CHECK and DEFAULT constraints maintain data integrity.

Simple Query: `SELECT * FROM Passenger WHERE Age > 20;`

Nested Query: `SELECT Name FROM Passenger WHERE Passenger_ID IN (SELECT Passenger_ID FROM Reservation);`

JOIN: `SELECT P.Name, R.Reservation_ID, R.Seat_Number FROM Passenger P JOIN Reservation R ON P.Passenger_ID = R.Passenger_ID;`

View: `CREATE VIEW Passenger_Reservation_View AS SELECT P.Name AS Passenger_Name, R.Reservation_ID, R.Seat_Number, R.Journey_Date FROM Passenger P JOIN Reservation R ON P.Passenger_ID = R.Passenger_ID;`

Trigger: A trigger can automatically record a new reservation in a Reservation_Audit table after an INSERT operation.

9. System Workflow

- Passenger
- Searches for Train
- Selects Train
- Makes Reservation
- Seat/Date Assigned
- Makes Payment
- Booking Confirmed

10. Advantages

- Reduces manual record keeping.
- Provides faster data retrieval.
- Reduces data duplication.
- Maintains relationships between railway records.
- Improves data consistency.
- Makes reservation information easier to access.
- Provides better organization of passenger and train data.
- Supports complex queries and reporting.

11. Limitations

- Real-time seat availability.
- Cancellation and refund processing.
- Multiple classes and quotas.
- Authentication and authorization.
- Live train tracking.
- Online payment gateway integration.
- Large-scale concurrent transactions.

12. Future Enhancements

- Online ticket cancellation
- Waiting-list management
- Automatic seat allocation
- Train schedule management
- User login and authentication
- Admin dashboard
- PNR generation
- SMS/email notifications
- Real-time train status
- Python GUI/web interface

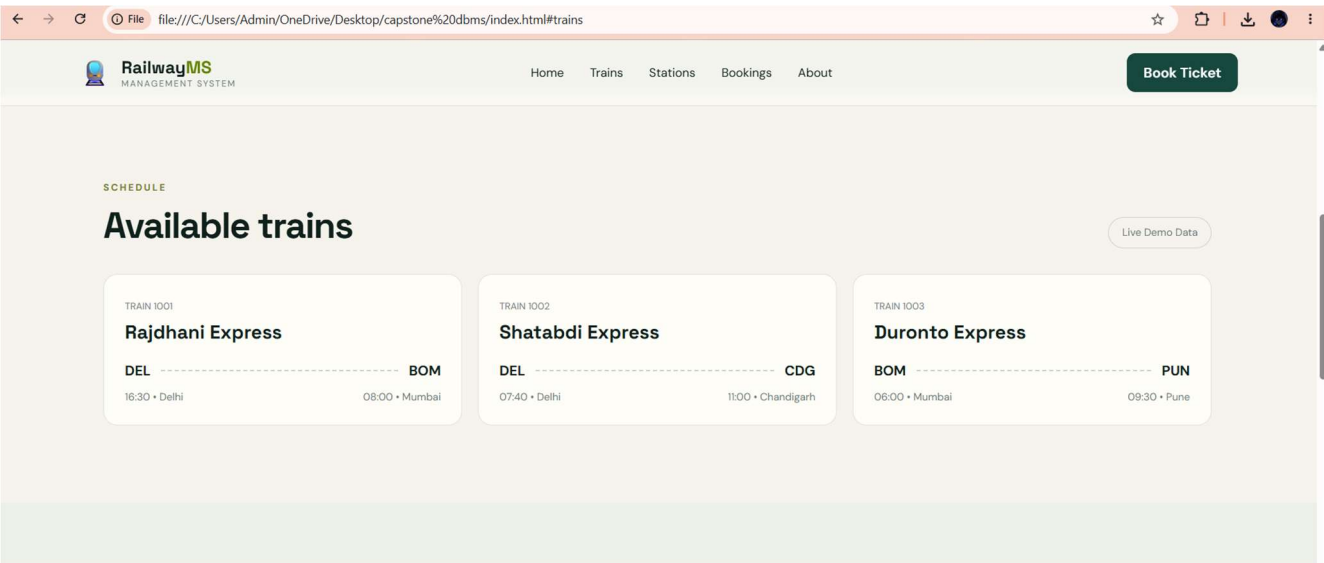
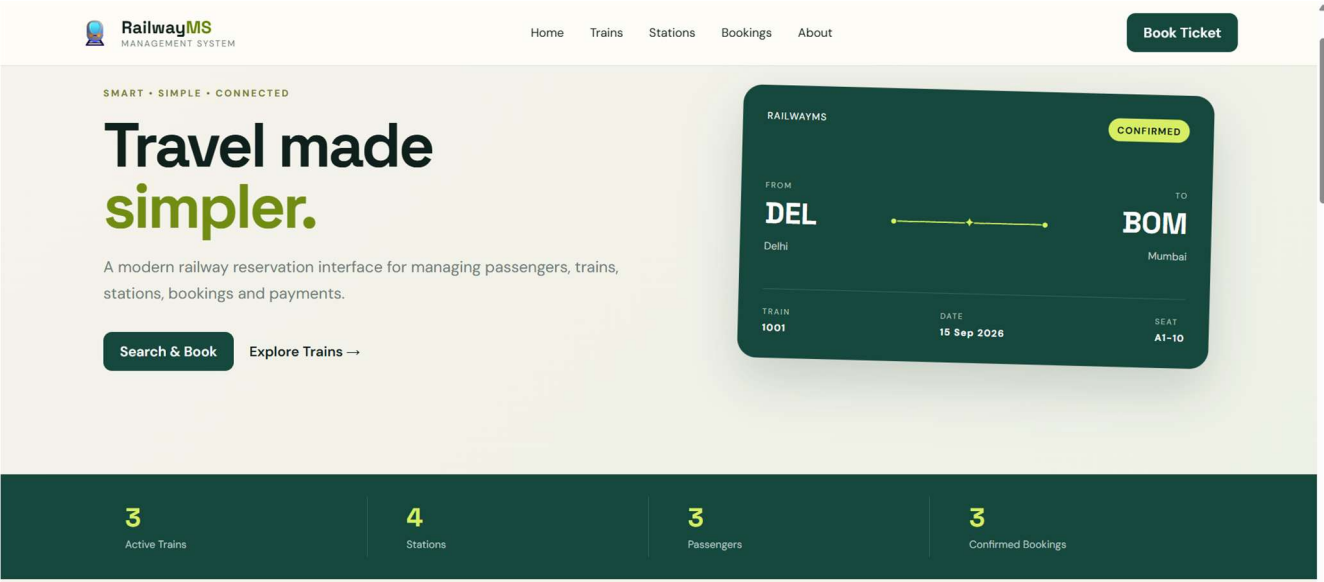
13. Conclusion

The Railway Management System demonstrates how a DBMS can be used to solve a real-world problem involving large amounts of interconnected data.

The project combines database design, ER/EER concepts, relational tables, DDL, DML, integrity constraints, simple queries, nested queries, joins, views and triggers into one practical application.

The system provides an organized way to manage passengers, trains, stations, reservations and payments while maintaining relationships and consistency among the stored data.

Thus, the Railway Management System serves as a practical DBMS capstone based on a real-life railway reservation scenario.



NETWORK

Major stations



New Delhi
Station 201 • Delhi



Mumbai Central
Station 202 • Mumbai



Chandigarh
Station 203 • Chandigarh



Pune Junction
Station 204 • Pune

RESERVATIONS

Recent bookings

PASSENGER	TRAIN	SEAT	JOURNEY DATE	STATUS	PAYMENT
Neerav Joshi	Rajdhani Express	Auto	15 Sept 2026	Confirmed	₹1,000 • Card
Rahul Sharma	Rajdhani Express	A1-10	15 Sep 2026	Confirmed	₹1,500 • UPI
Priya Mehta	Shatabdi Express	B2-15	18 Sep 2026	Confirmed	₹1,200 • Card
Aman Verma	Duronto Express	C1-08	20 Sep 2026	Confirmed	₹900 • UPI

NEW RESERVATION



Book your journey

Passenger Name

Muskaan Khoja

From

Chandigarh



To

Delhi



Journey Date

30-09-2026



Train

1002 — Shatabdi Express



Payment Mode

Net Banking



Confirm Reservation

Recent bookings

PASSENGER	TRAIN	SEAT	JOURNEY DATE	STATUS	PAYMENT
Vedant	Duronto Express	Auto	27 Nov 2026	Confirmed	₹1,000 • Card
Muskaan Khoja	Shatabdi Express	Auto	30 Sept 2026	Confirmed	₹1,000 • Net Banking
Neerav Joshi	Rajdhani Express	Auto	15 Sept 2026	Confirmed	₹1,000 • Card
Rahul Sharma	Rajdhani Express	A1-10	15 Sep 2026	Confirmed	₹1,500 • UPI
Priya Mehta	Shatabdi Express	B2-15	18 Sep 2026	Confirmed	₹1,200 • Card
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